
Noah Day

✉ noah.day@adelaide.edu.au 🌐 NoahDay 🔗 linkedin.com/in/NoahDay 🔗 https://noahday.github.io

EDUCATION

Doctor of Philosophy in Mathematical Sciences | *University of Adelaide* June 2021–Present
(submitted Feb 2025)

Modelling the Antarctic Marginal Ice Zone: Classification and Evolution in Response to Storms and Waves

- Numerical modelling of waves-in-ice in the Southern Ocean using state-of-the-art numerical sea ice models
- Novel statistical analysis to define the Antarctic marginal ice zone and comparison against satellite observations
- Quantification of the contribution of polar cyclones on the extent of the Antarctic marginal ice zone
- Development of a novel wave-ice framework for the Antarctic marginal ice zone

Honours Degree of Bachelor of Mathematical Sciences | *University of Adelaide* Feb 2017–Dec 2020
with First Class Honours

Honours GPA: 6.8/7

Honours thesis: *Stochastic Modelling of Vector-Borne Diseases*

PUBLICATIONS

Day, N. S., Bennetts, L. G., O'Farrell, S. P., Alberello, A., Montiel, F. (2024). Analysis of the Antarctic marginal ice zone based on unsupervised classification of standalone sea ice model data. *Journal of Geophysical Research: Oceans*, 129, e2024JC020953. <https://doi.org/10.1029/2024JC020953>.

Fraser, A., **Day, N. S.**, Wang, Z., Bennetts, L. G., Liu, Q., O'Farrell, S. P., Coleman, R., Voermans, J., Xu, S., Zhu, W., Auger, M., Massom, R., Wongpan, P., Craw, L., Brouwer, J., Toyota, T., Heil, P., Horvat, C. (2025). Revealing the Antarctic marginal ice zone: a decade-long wave-in-ice climatology. *In review at Nature Communications*.

Day, N. S., O'Farrell, S. P., Bennetts, L. G. (2025). The Short-Term Impacts of Winter Extratropical Cyclones on the Antarctic Ice Edge Location and Marginal Ice Zone Width. *In preparation*.

Day, N. S., O'Farrell, S. P., Alberello, A., Bennetts, L. G. (2025). Wave Propagation Through Broken and Consolidated Sea Ice Covers. *In preparation*.

EMPLOYMENT

Research Assistant | *University of Adelaide* March 2025–June 2025

Development and testing of coupled CICE6–WAVEWATCH III model configurations.

Teaching Assistant | *University of Adelaide* Aug 2021–Present

Practicals, workshops, tutorials, marking and exam invigilation for undergraduate and post-graduate mathematics courses.

- STATS 1000—Mathematics for Data Science I (S2 2024, S1 2023, S2 2022, S2 2021)
- STATS 7022—Data Science Post Graduate (S1 2024)
- MATHS 7027—Mathematical Foundations of Data Science (S1 2024, S2 2023)
- STATS 1000—Statistical Practice I (S1 2024)
- MATHS 1009—Introduction to Financial Mathematics (S1 2023)
- MATHS 1010—Applications of Quantitative Methods in Finance I (S2 2022)
- MATHS 1011—Mathematics IA (S1 2022)

ACHIEVEMENTS AND SKILLS

Achievements:

- Student oral presentation prize | *IGS International Symposium on Sea Ice (2023)*
- PhD top-up scholarship | Funded by the *Consortium for Ocean-Sea Ice Modelling in Australia* (2021–2024)
- Travel grants:
 - * DR Stranks Travelling Fellowship (\$3,640 AUD)
 - * University of Adelaide Graduate Research School Travel Grant (\$2,500 AUD)
 - * Quarterly Journal of Mechanics and Applied Mathematics (£1,300 GBP)
 - * CICE Workshop 2024 (\$2,920 USD)
 - * neXtSIM Workshop 2025 (\$2,138 USD)

Community Contributions:

- Hackathon Prize: Best pull request (PR) contribution to the *COSIMA Recipes* Github repository
- International model development workshop participation for the CICE and neXtSIM sea ice models

Languages: Python, MATLAB, Fortran, R, Julia

Tools: Git/GitHub, Unix Shell

Libraries: Xarray, pandas, NumPy, Matplotlib

CONFERENCES AND ORAL PRESENTATIONS

International:

- CICE Consortium User Workshop (2024)—Santa Fe, USA
- Los Alamos National Laboratory (2024)—Los Alamos, USA
- Ocean Sciences Meeting (2024)—New Orleans, USA
- University of Texas San Antonio Antarctic Sea Ice and Southern Ocean seminar series (2024)—Online
- UK Met Office polar group seminar (2023)—Exeter, UK
- International Symposium on Sea Ice (2023)—Bremerhaven, Germany
- EGU General Assembly (2023)—Vienna, Austria
- University of East Anglia School of Mathematics postgraduate student seminar series (2023)

Domestic:

- Consortium for Ocean-Sea Ice Modelling in Australia Weekly Meetings (2022, 2024, 2025)
- Ocean Modelling and Observations Workshop 2024—Canberra, Australia
- University of Adelaide School of Mathematics postgraduate student seminar series (2022, 2023, 2024)
- Consortium for Ocean-Sea Ice Modelling in Australia Workshop 2022—Hobart, Australia
- Institute for Marine and Antarctic Studies Sea Ice seminar series (2022)—Online

REFEREES

Professor Luke Bennetts | *PhD Principal Supervisor*
University of Melbourne, luke.bennetts@unimelb.edu.au

Dr Siobhan O'Farrell | *PhD Co-supervisor*
CSIRO Aspendale, siobhan.o'farrell@csiro.edu.au

Dr Andrew Black | *Honours Supervisor*
University of Adelaide, andrew.black@adelaide.edu.au